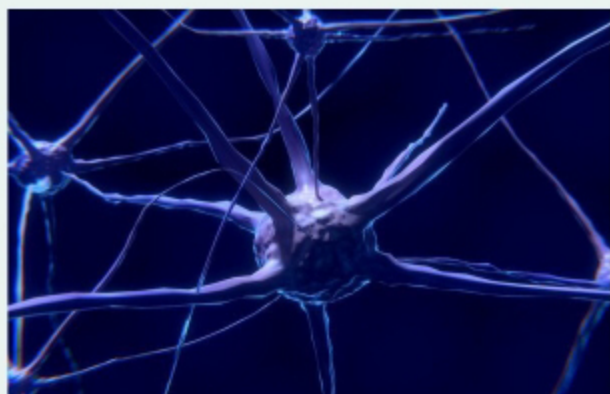


Open source software as accurate as humans in identifying active neurons

Tracking of active neurons has been made easier by the development of an open source software called CalmAn.

As reported by *Hi-Tech Chronicle*, the software enhances the analysis process using a combination of computational methods and machine learning techniques.



Dmitri Chklovskii, Eftychios Pnevmatikakis and Andrea Giovannucci at the Center for Computational Biology (Simons Foundation, New York), spearheaded the development of CalmAn with the aim of developing a tool that would help in tackling the enormous data sets produced by calcium imaging.

A journal published in January by the software developers showed CalmAn achieve near-human accuracy in locating active neurons based on calcium imaging data.

CalmAn has been in the market for a few years and has worked to benefit the calcium imaging community. There are over 100 labs that utilise the software. With the upgrades in the latest software, scientists can run it in a standard laptop and analyse data in real-time.

CalmAn adopts a unique technique that involves the addition of a special dye to brain tissue or neurons in a dish. The dye then binds to the calcium ions responsible for activating neurons. It is then viewed under ultraviolet rays whereby the calcium ion with the dye lights up, allowing the scientists or researchers to track the activity of the neuron visually.

Recent tests have shown that the software is nearly as accurate as humans in identifying active neurons.

Disaggregated Solution (PODS), we are contributing our capabilities and expertise to build future RANs on the foundation of virtualised network elements, white-box hardware, and open and standardised interfaces,” said Dr Badri Gomatam, group CTO, Sterlite Tech.

Once developed, the near real-time controller will help the Alliance operators accelerate the development and commercialisation of 5G networks globally, Gomatam added.

The company plans to work with the community to develop curated and hardened O-RAN based solutions. Its other plans include testing these solutions at Sterlite Networks, a fully operational carrier-neutral telecommunications network, and to establish field readiness in a carrier-grade environment.

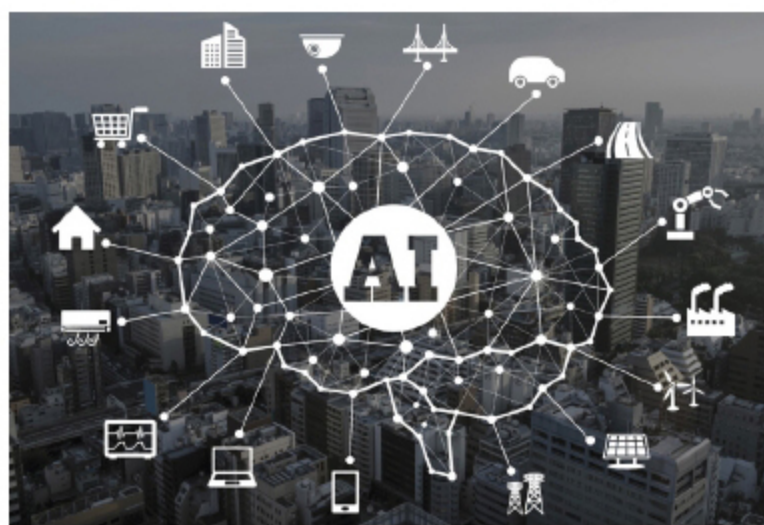
This unique approach will help Sterlite Tech’s carrier customers ascertain certification and conformance to the O-RAN Alliance specifications and accelerate their time to market.

Google open sources GPipe library to scale up deep neural network training

The Google AI research team has open sourced GPipe, a distributed machine learning library, for efficiently training large-scale deep neural network models under the Lingvo Framework.

GPipe uses synchronous stochastic gradient descent and pipeline parallelism for training. It allows researchers to easily deploy more accelerators to train larger models and to scale the performance without tuning hyper-parameters. “To enable efficient training across multiple accelerators, GPipe partitions a model across different accelerators and automatically splits a mini-batch of training examples into smaller micro-batches,” the team’s blog post stated.

The AI research team used GPipe to verify the hypothesis that scaling up existing neural networks can help achieve better model quality. The team trained an AmoebaNet-B with 557 million model parameters and an input image size of



480 x 480 pixels on the ImageNet ILSVRC-2012 data set. The model was able to reach 84.3 per cent top-1/97 per cent top-5 single-crop validation accuracy without the use of any external data.

The researchers also ran the transfer learning experiments on

the CIFAR10 and CIFAR100 data sets, when they observed that the giant models improved the best published CIFAR10 accuracy to 99 per cent and CIFAR100 accuracy to 91.3 per cent.

“The ongoing development and success of many practical machine learning applications, such as autonomous driving and medical imaging, depends on achieving the highest accuracy possible. As this often requires building larger and even more complex models, we are happy to provide GPipe to the broader research community, and hope it is a useful infrastructure for efficient training of large scale DNNs,” the team said.

Google AI researchers had also published a paper titled ‘GPipe: Efficient Training of Giant Neural Networks using Pipeline Parallelism’, in which they